Capstone Design

Week 4: Architecture, Wireframes, and Sprint 1 Kickoff

Deliverables Today

1. C4-lite architecture sketch
2. 3 low-fi wireframes
3. Design Doc v1
4. Sprint 1 GitHub issues

Why this week matters

- Ideas are cheap
- Build plans are harder
- Teams lose time when:
  - architecture is fuzzy
  - screens are unclear
  - tasks are not owned
- Today fixes that

What you should leave class with

- 1 architecture sketch
- 3 wireframes
- 1 design doc v1
- 4–8 Sprint 1 issues
- updated GitHub board

Today’s flow

1. Architecture: what the system includes
2. Wireframes: what the user sees
3. Design Doc v1: shared team reference
4. GitHub kickoff: turn plan into owned work

What “C4-lite” means in this class

C4-lite = a simple architecture sketch showing:

- who uses the system
- the main system
- major parts inside it
- external services
- how they connect

What to include, what to skip

Include:

- users
- frontend
- backend
- database
- external APIs/services
- main data flow

Skip

- deep framework internals
- fancy notation
- every class/function
- giant unreadable diagrams

Running example: Study Room Reservation App

Sample MVP:

- students can view available rooms
- students can reserve a room
- admin can manage room availability
- reservations are stored in a database

Example: System context view

Actors and systems:

- Student
- Admin
- Reservation Web App
- University Login (optional)
- Database
- Email/Notification service (optional)

Example: Container view

- Frontend
  - room list
  - booking form
  - reservation status
- Backend/API
  - room availability logic
  - booking logic
  - user/auth checks
- Database
  - rooms
  - reservations
  - users

Common architecture mistakes

- boxes with vague labels
- no user shown
- no external dependencies shown
- mixing product ideas with technical detail
- diagrams too big to read
- architecture that does not match the MVP

What a wireframe is

A wireframe is a:

- low-fidelity screen plan
- way to show layout and function
- fast tool for thinking, not decoration

Not:

- polished UI
- branding exercise
- final visual design

The 3-screen rule

For this week, make 3 low-fi wireframes max:

1. Entry / home screen
2. Core task screen
3. Result / detail / confirmation / admin screen

What a good wireframe includes

A good wireframe should show:

- screen title
- key buttons/actions
- main content areas
- labels for important sections
- arrows or notes for main user flow

Common wireframe mistakes

- too many screens
- too much visual polish too early
- unclear main action
- missing user flow
- no connection to MVP scope
- designing edge cases before the main case works

What Design Doc v1 is for

Design Doc v1 is your team's:

- shared reference
- build handoff
- scope control tool
- source for creating Sprint 1 issues

Required sections for Design Doc v1

Include:

- project purpose
- target users
- MVP scope
- C4-lite architecture
- 3 wireframes
- Sprint 1 build plan
- risks / assumptions
- scope cuts

From design doc to Sprint 1 issues

Turn the design doc into:

- feature issues
- setup / infrastructure issue
- testing / QA / accessibility issue
- blocker / risk issue if needed

What a good Sprint 1 issue looks like

A good issue has:

- clear task
- one owner
- reason it matters
- Definition of Done
- linked evidence when completed

Issue: Build room search page

- show available rooms
- filter by date/time
- display reserve button
- DoD: page works locally, PR opened, screenshots linked

GitHub board expectations

Your board should show:

- To Do
- Doing
- Done

And every card should have:

- owner
- clear status
- link to issue/PR
- Definition of Done

This week’s engineering practice: accessibility basics

Check for:

- clear headings
- labeled buttons/forms
- readable text/contrast
- sensible keyboard flow
- understandable screens

What gets submitted this week

One Weekly Sprint Packet per team, including:

- Design Doc v1 link
- architecture sketch
- 3 wireframes
- board snapshot or link
- Sprint 1 issue links
- 3-bullet demo script
- risks / blockers / scope cuts

Rule: Fill out the docs first. Link them in the Sprint Packet.

* Each student adds a comment with 2–3 contribution receipts

What good looks like this week

A strong Week 4 submission lets me answer:

- What are you building first?
- Who uses it?
- What are the main screens?
- What are the major system parts?
- What exactly is in Sprint 1?
- What are you intentionally not building yet?

Before you leave class

Before your team leaves today, you should have:

- architecture sketch done
- 3 wireframes done
- Design Doc v1 started or complete
- Sprint 1 issues created
- board updated
- next owner actions clear